KAIZEN MATHS

Lesson starter

Histograms and Frequency Density

Exam-Style Histograms

Name _____

SA1

Complete the frequency column using class width × frequency density.

Class interval	Class width	Frequency	Frequency density
$30 \leq x < 35$	5	?	2
$35 \leq x < 45$	10	?	3.50
$45 \leq x < 55$	10	?	1
$55 \leq x < 65$	10	?	2.50
$65 \leq x < 80$	15	?	3.50

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KAIZEN MATHS

Lesson starter

Histograms and Frequency Density

Exam-Style Histograms

Name _____

SB1

The table shows the distribution of height for plants. Estimate the median and interquartile range.

Class interval	Class width	Frequency	Frequency density
$20 \leq x < 35$	15	15	1
$35 \leq x < 50$	15	30	2
$50 \leq x < 60$	10	15	1.50
$60 \leq x < 80$	20	30	1.50
$80 \leq x < 100$	20	40	2

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Lesson starter

Histograms and Frequency Density

Exam-Style Histograms

Name _____

SC1

A class interval has width 15. Its bar on a histogram should represent 60 devices. Find the frequency density and explain how the bar should be drawn.

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KAIZEN MATHS

Lesson starter

Histograms and Frequency Density

Exam-Style Histograms

Name _____

SD1

Complete the frequency column using class width × frequency density.

Class interval	Class width	Frequency	Frequency density
$30 \leq x < 50$	20	?	1
$50 \leq x < 60$	10	?	2.50
$60 \leq x < 70$	10	?	3
$70 \leq x < 85$	15	?	1.50
$85 \leq x < 95$	10	?	2.50

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KAIZEN MATHS

Exit ticket

Histograms and Frequency Density

Exam-Style Histograms

Name _____

EA1

The table shows the distribution of mass for parcels. Estimate the median and interquartile range.

Class interval	Class width	Frequency	Frequency density
$30 \leq x < 35$	5	7.50	1.50
$35 \leq x < 50$	15	15	1
$50 \leq x < 60$	10	10	1
$60 \leq x < 65$	5	15	3
$65 \leq x < 85$	20	50	2.50

STA-FE00-L3 · EA

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Histograms and Frequency Density

Exam-Style Histograms

Name _____

EB1

A class interval has width 20. Its bar on a histogram should represent 40 students. Find the frequency density and explain how the bar should be drawn.

STA-FE00-L3 · EB

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Histograms and Frequency Density

Exam-Style Histograms

Name _____

EC1

Complete the frequency column using class width × frequency density.

Class interval	Class width	Frequency	Frequency density
$40 \leq x < 55$	15	?	2
$55 \leq x < 75$	20	?	2.50
$75 \leq x < 85$	10	?	1.50
$85 \leq x < 105$	20	?	4
$105 \leq x < 110$	5	?	4

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Histograms and Frequency Density

Exam-Style Histograms

Name _____

EA1

The table shows the distribution of mass for parcels. Estimate the median and interquartile range.

Class interval	Class width	Frequency	Frequency density
$30 \leq x < 35$	5	7.50	1.50
$35 \leq x < 50$	15	15	1
$50 \leq x < 60$	10	10	1
$60 \leq x < 65$	5	15	3
$65 \leq x < 85$	20	50	2.50

STA-FE00-L3 · EA

Show your working

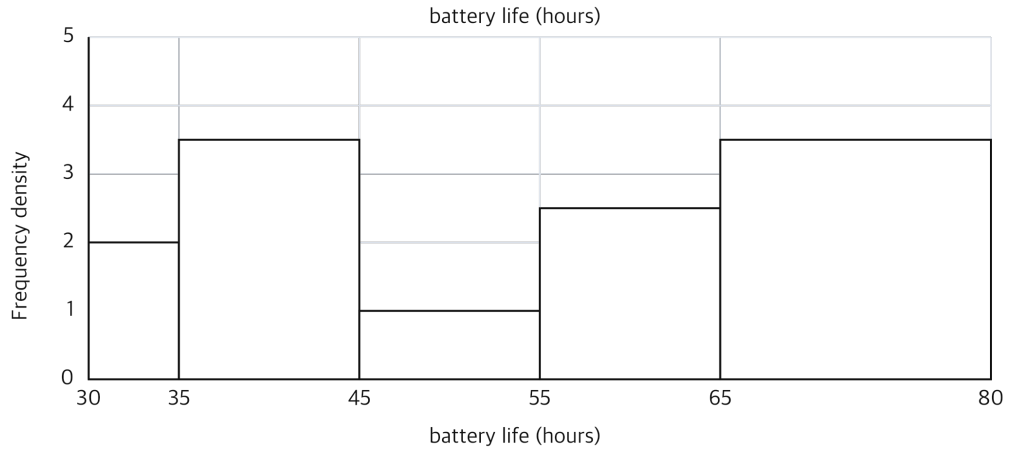
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Histograms and Frequency Density · Exam-Style Histograms

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SA1

Use the histogram to complete the frequency column in the table.



Class interval	Class width	Frequency	Frequency density
$30 \leq x < 35$	5	?	2
$35 \leq x < 45$	10	?	3.50
$45 \leq x < 55$	10	?	1
$55 \leq x < 65$	10	?	2.50
$65 \leq x < 80$	15	?	3.50

Answer:

$30 \leq x < 35$: 10; $35 \leq x < 45$: 35; $45 \leq x < 55$: 10; $55 \leq x < 65$: 25; $65 \leq x < 80$: 52.50

Working:

Read the height of each bar as the frequency density.

Work out the class width for each class.

frequency = class width × frequency density

For 30 ≤ x < 35, class width = 35 - 30 = 5.

frequency = 5 × 2 = 10

For 35 ≤ x < 45, class width = 45 - 35 = 10.

frequency = 10 × 3.50 = 35

For 45 ≤ x < 55, class width = 55 - 45 = 10.

frequency = 10 × 1 = 10

For 55 ≤ x < 65, class width = 65 - 55 = 10.

frequency = 10 × 2.50 = 25

For 65 ≤ x < 80, class width = 80 - 65 = 15.

frequency = 15 × 3.50 = 52.50

$30 \leq x < 35$: 10; $35 \leq x < 45$: 35; $45 \leq x < 55$: 10; $55 \leq x < 65$: 25; $65 \leq x < 80$: 52.50

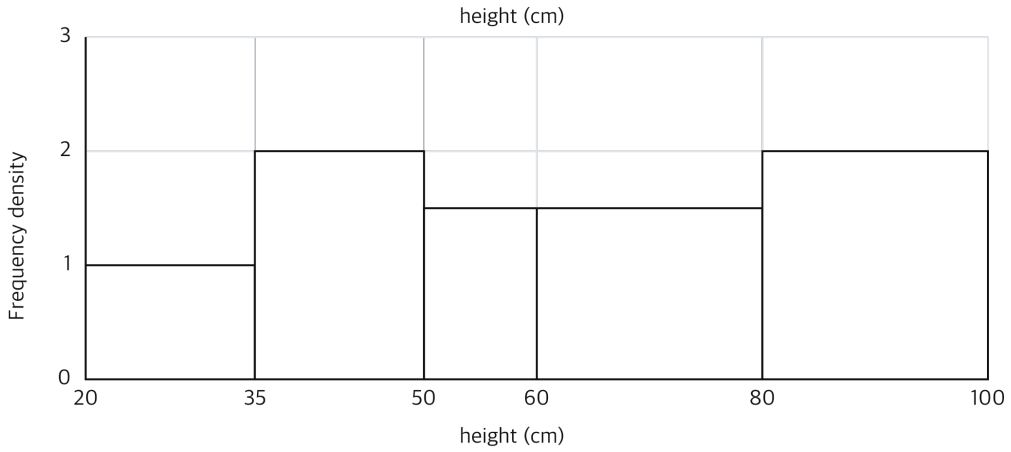
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Histograms and Frequency Density · Exam-Style Histograms

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SB1

The histogram and table show the distribution of height for plants. Estimate the median and interquartile range.



Class interval	Class width	Frequency	Frequency density
$20 \leq x < 35$	15	15	1
$35 \leq x < 50$	15	30	2
$50 \leq x < 60$	10	15	1.50
$60 \leq x < 80$	20	30	1.50
$80 \leq x < 100$	20	40	2

Answer:

median ≈ 63.3 cm; IQR ≈ 40 cm

Working:

First find the total frequency.

total frequency = 15 + 30 + 15 + 30 + 40 = 130

Find the positions of the quartiles and median.

Q1 position = 130 ÷ 4 = 32.50

median position = 130 ÷ 2 = 65

Q3 position = 3 × 130 ÷ 4 = 97.50

Locate the lower quartile. It lies in 35 ≤ x < 50 because the cumulative frequency before the class is 15.

Frequency needed inside the class = 32.50 - 15 = 17.50.

Distance across the class = frequency needed ÷ frequency density.

distance = 17.50 ÷ 2 = 8.75

lower quartile ≈ 35 + 8.75 = 43.8 cm

Locate the median. It lies in 60 ≤ x < 80 because the cumulative frequency before the class is 60.

Frequency needed inside the class = 65 - 60 = 5.

Distance across the class = frequency needed ÷ frequency density.

distance = 5 ÷ 1.50 = 3.33

Teacher answers & working

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SB1 (continued)

median ≈ 60 + 3.33 = 63.3 cm

Locate the upper quartile. It lies in 80 ≤ x < 100 because the cumulative frequency before the class is 90.

Frequency needed inside the class = 97.50 - 90 = 7.50.

Distance across the class = frequency needed ÷ frequency density.

distance = 7.50 ÷ 2 = 3.75

upper quartile ≈ 80 + 3.75 = 83.8 cm

IQR ≈ 83.8 - 43.8 = 40 cm

median ≈ 63.3 cm; IQR ≈ 40 cm

SC1

A class interval has width 15. Its bar on a histogram should represent 60 devices. Find the frequency density and explain how the bar should be drawn.

Answer:

frequency density = 4; draw the bar over width 15 with height 4

Working:

Use frequency density = frequency ÷ class width.

frequency density = 60 ÷ 15 = 4

The width of the bar on the horizontal axis is the class width.

The height of the bar on the vertical axis is the frequency density.

draw a bar of width 15 and height 4

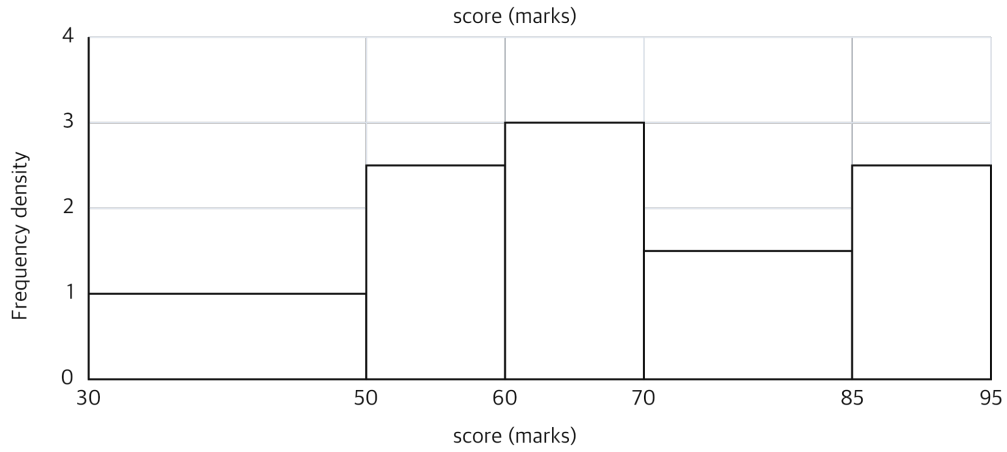
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Histograms and Frequency Density · Exam-Style Histograms

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SD1

Use the histogram to complete the frequency column in the table.



Class interval	Class width	Frequency	Frequency density
$30 \leq x < 50$	20	?	1
$50 \leq x < 60$	10	?	2.50
$60 \leq x < 70$	10	?	3
$70 \leq x < 85$	15	?	1.50
$85 \leq x < 95$	10	?	2.50

Answer:

$30 \leq x < 50$: 20; $50 \leq x < 60$: 25; $60 \leq x < 70$: 30; $70 \leq x < 85$: 22.50; $85 \leq x < 95$: 25

Working:

Read the height of each bar as the frequency density.

Work out the class width for each class.

frequency = class width × frequency density

For 30 ≤ x < 50, class width = 50 - 30 = 20.

frequency = 20 × 1 = 20

For 50 ≤ x < 60, class width = 60 - 50 = 10.

frequency = 10 × 2.50 = 25

For 60 ≤ x < 70, class width = 70 - 60 = 10.

frequency = 10 × 3 = 30

For 70 ≤ x < 85, class width = 85 - 70 = 15.

frequency = 15 × 1.50 = 22.50

For 85 ≤ x < 95, class width = 95 - 85 = 10.

frequency = 10 × 2.50 = 25

$30 \leq x < 50$: 20; $50 \leq x < 60$: 25; $60 \leq x < 70$: 30; $70 \leq x < 85$: 22.50; $85 \leq x < 95$: 25

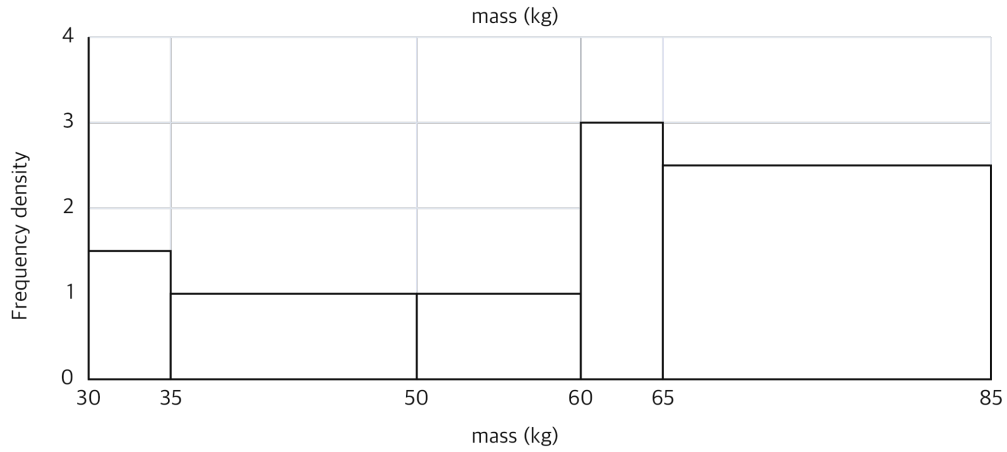
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Histograms and Frequency Density · Exam-Style Histograms

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EA1

The histogram and table show the distribution of mass for parcels. Estimate the median and interquartile range.



Class interval	Class width	Frequency	Frequency density
$30 \leq x < 35$	5	7.50	1.50
$35 \leq x < 50$	15	15	1
$50 \leq x < 60$	10	10	1
$60 \leq x < 65$	5	15	3
$65 \leq x < 85$	20	50	2.50

Answer:

median ≈ 65.5 kg; IQR ≈ 23.4 kg

Working:

First find the total frequency.

total frequency = 7.50 + 15 + 10 + 15 + 50 = 97.50

Find the positions of the quartiles and median.

Q1 position = 97.50 ÷ 4 = 24.38

median position = 97.50 ÷ 2 = 48.75

Q3 position = 3 × 97.50 ÷ 4 = 73.13

Locate the lower quartile. It lies in 50 ≤ x < 60 because the cumulative frequency before the class is 22.50.

Frequency needed inside the class = 24.38 - 22.50 = 1.88.

Distance across the class = frequency needed ÷ frequency density.

distance = 1.88 ÷ 1 = 1.88

lower quartile ≈ 50 + 1.88 = 51.9 kg

Locate the median. It lies in 65 ≤ x < 85 because the cumulative frequency before the class is 47.50.

Frequency needed inside the class = 48.75 - 47.50 = 1.25.

Distance across the class = frequency needed ÷ frequency density.

distance = 1.25 ÷ 2.50 = 0.50

Teacher answers & working

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EA1 (continued)

median ≈ 65 + 0.50 = 65.5 kg

Locate the upper quartile. It lies in 65 ≤ x < 85 because the cumulative frequency before the class is 47.50.

Frequency needed inside the class = 73.13 - 47.50 = 25.63.

Distance across the class = frequency needed ÷ frequency density.

distance = 25.63 ÷ 2.50 = 10.25

upper quartile ≈ 65 + 10.25 = 75.3 kg

IQR ≈ 75.3 - 51.9 = 23.4 kg

median ≈ 65.5 kg; IQR ≈ 23.4 kg

EB1

A class interval has width 20. Its bar on a histogram should represent 40 students. Find the frequency density and explain how the bar should be drawn.

Answer:

frequency density = 2; draw the bar over width 20 with height 2

Working:

Use frequency density = frequency ÷ class width.

frequency density = 40 ÷ 20 = 2

The width of the bar on the horizontal axis is the class width.

The height of the bar on the vertical axis is the frequency density.

draw a bar of width 20 and height 2

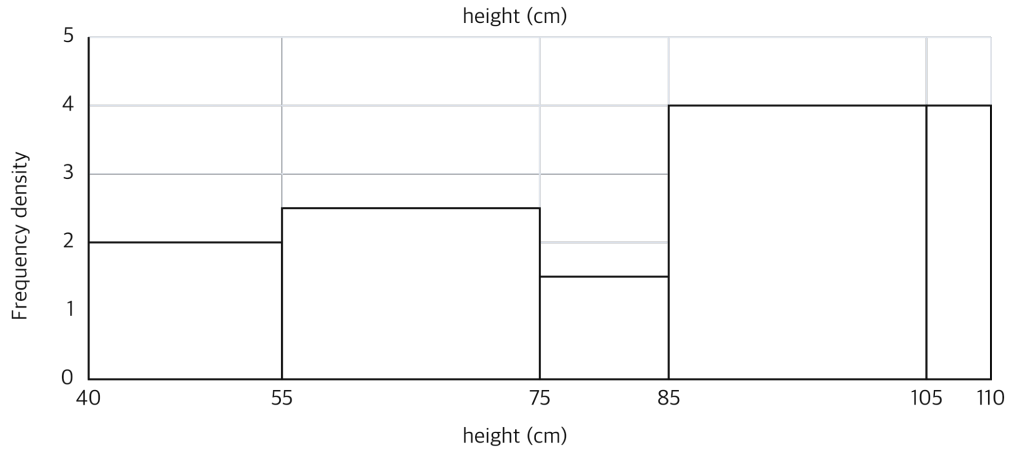
Teacher answers & working

Histograms and Frequency Density · Exam-Style Histograms

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EC1

Use the histogram to complete the frequency column in the table.



Class interval	Class width	Frequency	Frequency density
$40 \leq x < 55$	15	?	2
$55 \leq x < 75$	20	?	2.50
$75 \leq x < 85$	10	?	1.50
$85 \leq x < 105$	20	?	4
$105 \leq x < 110$	5	?	4

Answer:

$40 \leq x < 55$: 30; $55 \leq x < 75$: 50; $75 \leq x < 85$: 15; $85 \leq x < 105$: 80; $105 \leq x < 110$: 20

Working:

Read the height of each bar as the frequency density.

Work out the class width for each class.

frequency = class width × frequency density

For 40 ≤ x < 55, class width = 55 - 40 = 15.

frequency = 15 × 2 = 30

For 55 ≤ x < 75, class width = 75 - 55 = 20.

frequency = 20 × 2.50 = 50

For 75 ≤ x < 85, class width = 85 - 75 = 10.

frequency = 10 × 1.50 = 15

For 85 ≤ x < 105, class width = 105 - 85 = 20.

frequency = 20 × 4 = 80

For 105 ≤ x < 110, class width = 110 - 105 = 5.

frequency = 5 × 4 = 20

$40 \leq x < 55$: 30; $55 \leq x < 75$: 50; $75 \leq x < 85$: 15; $85 \leq x < 105$: 80; $105 \leq x < 110$: 20